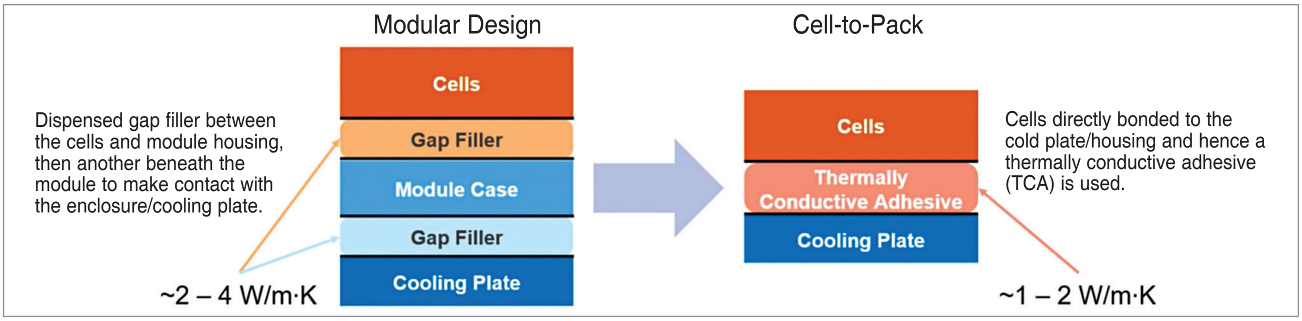

Fig. 2: EV battery pack designs (Credit: IDTechEx)

electronics industry, TIMs are crucial for managing heat dissipation in various applications, from electric vehicles to power electronics. TIM fillers play a significant role in enhancing the thermal performance and reliability of electronic devices.

Copper sintering is becoming increasingly popular as a cost-effective alternative to silver sintering, showing significant promise for widespread adoption, especially in the automotive sector. Meanwhile, boron nitride fillers, known for providing high thermal conductivity with minimal impact on electrical insulation, face challenges in achieving widespread use due to their inflated cost.

Whether in EV power electronics, data centres, 5G infrastructure, or ADAS, the choice and performance of TIMs are critical for ensuring efficient thermal management. As the industry progresses, innovations in TIM technology will continue to play a pivotal role in developing and enhancing electronic systems. The table on previous page shows the pros and cons of different TIM fillers to understand their suitability for various applications:

Applications in electric vehicles

Battery pack industry. TIMs are crucial for managing heat in EV battery packs. Traditional modular designs use multiple layers of TIMs between cells and cooling plates. However, the shift to cell-to-pack designs reduces the number of thermal interfaces from two to one, potentially lowering thermal resistance and improving performance. This transition also emphasises the need for TIMs with mechanical strength to bond cells to cooling plates.

Power electronics. The transition from silicon IGBT to silicon carbide MOSFETs in EV power electronics has significantly reduced die area, decreasing the amount of die-attach materials required. However, the higher junction temperatures and increased heat flux of silicon carbide MOSFETs necessitate more advanced and expensive die-attach materials, such as silver sintering and future copper sintering technologies.

The choice of TIM fillers depends on specific application requirements, balancing factors such as cost, thermal and electrical conductivity, and mechanical properties. As the electronics industry evolves, demand for more efficient and multifunctional TIMs will drive further innovation and development in filler materials.

As electric vehicles continue to gain momentum, the importance of thermal management in power electronics cannot be overstated. Effective thermal management ensures optimal performance, longevity, and safety of EV components. Central to this are TIMs, broadly classified into two categories: TIM 1 and TIM 2, with a transitional category sometimes referred to as TIM 1.5.

A notable trend is the shift by many automotive OEMs from traditional base plate configurations to direct bonding of the direct bonded copper (DBC) substrate onto the heatsink. This creates a TIM 1.5 category, where the sintering layer between the DBC and the heatsink is considered TIM 1.5. This evolution highlights the industry's move towards more efficient and integrated thermal management solutions.

How crucial are TIM properties to elevating EV power electronics performance?

The area per kilowatt is a crucial metric when evaluating TIM 2 in EV power electronics. Evaluating this metric is essential for advancing EV technology and meeting the demands of high-performance, cost-effective electric transportation solutions. Key takeaways include:

- Achieving a low TIM 2 area per kilowatt is critical for high-performance systems
- High-performance TIMs often come with a higher cost
- The choice of TIM involves balancing weight, cost, area, and thermal conductivity
- In most car models, the TIM area per kilowatt is less than 200mm^2

Therefore, careful consideration of TIM properties and their impact on overall system performance remains a key focus in the design and enhancement of EV power electronics.

Expansion into data centres and 5G infrastructure

Beyond EVs, TIMs play a significant role in data centres and 5G